
Quantized Small Language Models as Tool-Using Agents: Memory Savings for Free, Fine-Tuning with Care

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Abstract

1 Agentic AI systems typically rely on large language models whose memory foot-
2 print prevents deployment on constrained hardware. We present a systematic em-
3 pirical study of whether a *small* language model (Phi-3-mini, 3.8B parameters)
4 remains a capable tool-using agent under NF4 4-bit quantization, and whether
5 lightweight QLoRA fine-tuning on generic agent demonstrations helps or hurts.
6 Using a single unified ReAct evaluation harness across three benchmarks—a 50-
7 task diagnostic agentic suite, GSM8K with a calculator tool, and HotpotQA (dis-
8 tractor) with a search tool—we find that INT4 quantization is essentially free for
9 agentic capability. On the diagnostic suite the FP16 and INT4 agents succeed
10 and fail on *exactly the same tasks* (zero discordant pairs, $n=50$); on the stan-
11 dard benchmarks paired McNemar tests find no significant difference ($p=0.58$
12 on GSM8K, $p=0.63$ on HotpotQA), with weight memory falling from 7.64 GB
13 to 2.21 GB (71%). We document two deployment-relevant caveats. First, on a
14 T4, INT4 inference is roughly $2\times$ *slower* than FP16 because dequantization over-
15 head dominates at batch size one, so quantization buys memory, not speed, on
16 commodity accelerators. Second, QLoRA fine-tuning on 15 short generic Re-
17 Act demonstrations (0.25% of parameters, under two minutes of training) leaves
18 diagnostic-suite accuracy unchanged but degrades performance on the standard
19 benchmarks (pooled McNemar $p=0.041$), concentrated on multi-hop question an-
20 swering. Trace analysis identifies the mechanism unambiguously: the fine-tuned
21 agent issues a second search on 10 of 100 questions where the base INT4 agent
22 issues one on 39 (Wilcoxon signed-rank $p<10^{-5}$), and second searches are pre-
23 cisely the ones that pay off. Our results suggest quantized SLM agents are viable
24 for deployment, but that naive behavioral fine-tuning on short traces can silently
25 truncate exploratory behavior in a way that per-benchmark accuracy alone does
26 not reveal. Code, evaluation harness, per-task traces, and all raw results are re-
27 leased.

1 Introduction

29 Agentic AI systems—language models that decompose tasks, invoke external tools, and act over
30 multiple steps—have advanced rapidly, but almost exclusively on top of large proprietary models.
31 This creates a deployment gap: the settings where autonomous agents are most valuable (on-device
32 assistants, on-premise enterprise automation, low-cost cloud inference) are precisely those where
33 serving a 70B+ parameter model is infeasible. Recent position work argues that small language
34 models (SLMs) are the natural substrate for agentic AI [1], since agent workloads are dominated by
35 narrow, repetitive tool-calling rather than open-ended generation.

Two efficiency levers make SLM agents plausible: post-training quantization [5, 7], which shrinks memory at some risk to capability, and parameter-efficient fine-tuning (PEFT) [5, 6], which can specialize behavior at negligible cost. What is missing is a careful account of how these levers interact with *agentic* behavior specifically: multi-step reasoning, tool selection, and iterative search are plausibly more fragile under quantization than single-turn generation, and behavioral fine-tuning may have out-of-distribution side effects that single-metric evaluations miss.

This paper contributes such an account for one representative open SLM, Phi-3-mini (3.8B) [3], evaluated end-to-end as a ReAct agent [4] on free commodity hardware (a single NVIDIA T4). We ask three questions. **(Q1)** How much agentic capability is lost at INT4 precision? **(Q2)** What are the true latency and memory trade-offs on commodity GPUs? **(Q3)** Does lightweight QLoRA fine-tuning on a handful of generic agent demonstrations improve agentic behavior, and does any change transfer to standard benchmarks?

Because every configuration runs the same task list in the same order under a single harness, our results are *paired*. We therefore use McNemar’s exact test for accuracy and Wilcoxon signed-rank for tool-call depth throughout, which is substantially more informative at $n=100$ than comparing independent proportions. Our findings:

1. **INT4 quantization is essentially free for agentic capability.** On the diagnostic suite, FP16 and INT4 produce *identical* per-task outcomes—zero discordant pairs across 50 tasks. On GSM8K and HotpotQA, paired tests find no significant difference ($p=0.58$, $p=0.63$), and the paired confidence intervals bound any INT4 penalty at roughly 11 accuracy points. Weight memory falls 71% (7.64 GB $\rightarrow$ 2.21 GB) (Q1).
2. **Quantization buys memory, not speed, on commodity hardware.** End-to-end task latency at INT4 is roughly $2\times$ FP16 on a T4 (6.96 s vs. 3.59 s per task on our diagnostic suite), because dequantization overhead dominates when compute, not memory bandwidth, is the bottleneck (Q2).
3. **Fine-tune with care: short generic traces suppress follow-up search.** QLoRA on 15 short ReAct demonstrations leaves diagnostic-suite accuracy unchanged (98.0%) but degrades the standard benchmarks (pooled McNemar $p=0.041$), with the effect concentrated on HotpotQA. The behavioral signal is far stronger than the accuracy signal: mean executed searches per question fall from 1.48 to 1.13, and the proportion of questions receiving a second search collapses from 39% to 10% (Wilcoxon signed-rank $p=8.8 \times 10^{-6}$, rank-biserial $r=0.83$) (Q3).

We view the third finding—a well-diagnosed negative result—as the most actionable: it identifies a concrete failure mode that any practitioner fine-tuning small agent models on demonstration traces should measure, and that per-benchmark accuracy alone does not reveal.

2 Related Work

Small language models. Capable sub-10B models—Phi-3 [3], Gemma [16], Mistral-7B [17], Qwen2 [18]—have narrowed the gap to much larger models, and surveys [19, 20] catalogue the design space. Belcak and Heinrich [1] argue that SLMs will dominate agentic deployments; Srivastava et al. [2] build a framework enabling SLMs as autonomous agents. Our work is complementary: rather than proposing a framework, we measure how the two standard efficiency levers affect agentic behavior under a minimal scaffold.

Quantization and parameter-efficient fine-tuning. Post-training quantization to 8-bit [7] and 4-bit [5, 8] preserves perplexity and single-turn accuracy remarkably well, and LoRA [6] and QLoRA [5] enable fine-tuning with a fraction of a percent of trainable parameters; distillation offers a complementary route [9]. Most quantization evaluations measure static benchmarks; we measure closed-loop agent behavior, where errors compound across steps and the number of tool invocations is itself an outcome.

Language agents and benchmarks. ReAct [4] interleaves reasoning traces with tool calls and remains the canonical minimal agent scaffold; Toolformer [10] showed models can learn API use self-supervised. Agent benchmarks such as AgentBench [11], ALFWorld [12] and WebArena [13]

target large models and heavy environments, and are poorly matched to 3.8B models on free hardware where format compliance dominates the score. We therefore combine a controlled diagnostic suite with GSM8K [14] (calculator tool) and HotpotQA [15] in the distractor setting (search tool).

3 Method

3.1 Agent architecture

We implement a minimal ReAct agent. The model receives a system prompt specifying the available tools and the Thought/Action/Observation format, and generates greedily (temperature 0, `do_sample=False`). A stopping criterion halts generation whenever the model emits `Observation::`; the named tool is then *actually executed*, its real output is injected into the context, and generation resumes. This guarantees observations are never hallucinated. The loop ends when the model emits `Final Answer:` or a step budget is reached (6 steps; 8 for GSM8K). Two implementation details proved necessary for small models: (i) a minimum-new-tokens guard (20 tokens) before the stopping criterion is armed, preventing spurious first-token stops, and (ii) trace trimming that truncates generation drift (e.g., fabricated follow-up questions) at known markers.

If no `Final Answer:` span is produced, the answer field falls back to the last line of the trace. We audited this fallback across all 750 agent runs: 21 answers fell back to observation text, and none of them scored correct, so the fallback does not inflate any reported accuracy.

3.2 Tools

`calculator[expr]` evaluates sanitized arithmetic expressions (rejecting any input outside a numeric-and-operator character class). `search[query]` retrieves from a benchmark-specific source: a fixed 27-entry fact base for the diagnostic suite, and, for HotpotQA, the 10 paragraphs (2 gold, 8 distractors) provided with each question, ranked by title-weighted lexical overlap, returning the single highest-scoring paragraph truncated to 600 characters. Because only one paragraph is returned per call, any question requiring both gold paragraphs necessarily requires at least two search calls—a fact we rely on in Section 4.3.

3.3 Benchmarks

Custom-50 (diagnostic). 50 tasks in five categories of 10: *single_lookup*, *arithmetic*, *multi_step* (search then calculate), *tool_selection* (must choose the correct tool), and *sequential* (explicit multi-action instructions). Scoring is substring match of the expected value in the final answer. This suite is a *format-compliance and tool-routing check*, not a capability measure: all configurations saturate it (Section 4.1), so it discriminates only in the sense of confirming that the agent loop functions identically across precisions. **GSM8K.** 100 problems sampled (seed 42) from the test split; calculator tool only; one worked example in the system prompt; scoring is exact numeric match against the gold answer. **HotpotQA (distractor).** 100 validation questions with span answers (yes/no questions excluded to keep inclusion-match scoring unambiguous), sampled with seed 42; scoring is normalized-answer inclusion (lowercasing, punctuation and article removal).

3.4 Configurations

FP16: the unmodified model in half precision (7.64 GB weight memory). **INT4:** NF4 double quantization via bitsandbytes [5] with FP16 compute (2.21 GB). **INT4+QLoRA:** the INT4 model with LoRA adapters ($r=16$, $\alpha=32$) on the attention projections (`qkv_proj`, `o_proj`), 9,437,184 trainable parameters (0.2464%), trained for 8 epochs at learning rate 2×10^{-4} (cosine schedule, warmup ratio 0.1, effective batch size 4, paged 8-bit AdamW) on **15 hand-written generic ReAct demonstrations** covering the five task patterns. The demonstrations average 1.60 tool calls each and are fully disjoint from all evaluation tasks in entities, numbers, and wording; *no GSM8K or HotpotQA data is used*. Training takes 1.8 minutes on a single T4. All runs use one T4 GPU, greedy decoding, seed 42, and identical harness code; the only variable is the model configuration. Environment: transformers 4.41.2, torch 2.10.0+cu128, datasets 2.19.1, peft 0.11.1, trl 0.8.6, accelerate 0.30.1, CUDA 12.8, Python 3.12, NVIDIA Tesla T4. The bitsandbytes version for the replication and trace-depth analysis of Section 4.4 is 0.50.1; for the original evaluation round reported in Table 1 it was installed

Table 1: Standard benchmarks (100 tasks each, seed 42). The agent is *not* fine-tuned on either dataset. Tool calls are the mean number of executed tool invocations per task. p -values are exact McNemar against the INT4 row within the same benchmark.

Benchmark	Config	Accuracy (%)	Latency (s)	Tool calls	p vs. INT4
GSM8K	FP16	71.0	7.32	3.38	0.581
	INT4	68.0	12.31	3.22	—
	INT4+QLoRA	65.0	14.31	3.25	0.388
HotpotQA (distractor)	FP16	47.0	5.46	1.65	0.629
	INT4	44.0	9.03	1.47	—
	INT4+QLoRA	38.0	7.30	1.13	0.070

as ≥ 0.46 . 1 and the resolved version was not recorded. Given our own findings in Section 4.5 we flag this as a gap rather than omit it; Section 4.4 quantifies how much it matters in practice.

4 Results

4.1 Q1: Quantization is essentially free for agentic capability

On the diagnostic suite all three configurations score 98.0% (49/50), with the single failure in *multi_step* and every other category at 100%. The paired view is stronger than the aggregate: FP16 and INT4 have **zero discordant pairs**—they succeed and fail on exactly the same 50 tasks. This is an equivalence result on this suite, not merely an undetected difference. Mean latency is 3.59 s (FP16), 6.96 s (INT4) and 7.37 s (INT4+QLoRA).

Table 1 shows the standard benchmarks. INT4 loses 3 points on GSM8K (71%→68%) and 3 points on HotpotQA (47%→44%). Paired McNemar tests find no significant difference in either case (GSM8K: 8 vs. 5 discordant, $p=0.58$; HotpotQA: 10 vs. 7 discordant, $p=0.63$). We state the result as a bound rather than a null: the paired 95% confidence intervals are $[-4.3, +10.8]$ on GSM8K and $[-5.1, +11.1]$ on HotpotQA, so our design rules out an INT4 penalty larger than roughly 11 points but cannot resolve differences below that. Within that bound, quantization costs nothing measurable while removing 71% of weight memory.

4.2 Q2: On a T4, quantization buys memory, not speed

Across every benchmark, FP16 is the fastest configuration—roughly $2\times$ faster than INT4 per task—despite INT4’s smaller weights. On a T4, NF4 weights must be dequantized to FP16 for every matrix multiplication, and this overhead dominates at batch size 1, where the GPU is not memory-bandwidth bound. The T4 is a Turing-generation accelerator with no native low-precision matmul path for NF4; this result is a property of that hardware class and should not be read as a general claim about quantized inference.

The practical implication is a deployment decision rule: choose INT4 when the constraint is *fitting* the model (edge devices, multi-tenant serving, freeing VRAM for context), and FP16 when the constraint is *latency* and the model already fits. Reported speedups from quantization on server-class GPUs with fused low-bit kernels do not automatically transfer to commodity accelerators, and end-to-end agent latency—which compounds per-step generation over multiple tool calls—amplifies the difference.

We report weight memory (7.64 GB vs. 2.21 GB) as measured by `get_memory_footprint()`. This excludes activations and the KV cache; peak allocated memory during evaluation is the quantity that determines whether a model fits on a given device, and our harness now records it alongside every run.

4.3 Q3: Generic-trace QLoRA suppresses follow-up search

QLoRA fine-tuning leaves the diagnostic suite unchanged (98.0%). On the standard benchmarks it costs 3 points on GSM8K and 6 points on HotpotQA. Neither benchmark alone reaches significance (GSM8K $p=0.39$; HotpotQA $p=0.070$), but the effect is directionally consistent across both, and

Table 2: Distribution of executed tool calls per HotpotQA question (replication run, $n=100$ paired). Our search tool returns one paragraph per call, so any question needing both gold paragraphs requires at least two calls. Fine-tuning removes three quarters of the second searches.

Config	0 calls	1 call	2 calls	3+ calls	≥ 2 calls
INT4	1	60	35	4	39
INT4+QLoRA	1	89	9	1	10

pooling the discordant pairs over the two benchmarks gives 15 vs. 5, exact McNemar $p=0.041$. The accuracy evidence is therefore suggestive but underpowered at $n=100$, and we do not rest the claim on it.

The decisive evidence is behavioral. Table 2 gives the distribution of executed tool calls per HotpotQA question. The base INT4 agent issues a second search on 39 of 100 questions; after fine-tuning it does so on 10. Across the 100 paired questions the fine-tuned agent used fewer calls on 30 and more on 2 (paired Wilcoxon signed-rank $W=44$, $p=8.8 \times 10^{-6}$, rank-biserial $r=0.83$).

Second searches are not idle. In both configurations, questions that received at least two calls scored substantially higher than those that received one (INT4: 51.3% vs. 37.7%; INT4+QLoRA: 60.0% vs. 35.6%). Fine-tuning therefore suppresses precisely the behavior that carries the multi-hop questions. Consistent with this, fine-tuned HotpotQA latency *drops* (8.24 s $\rightarrow$ 6.51 s): the agent is faster because it gives up earlier.

Two observations constrain the mechanism, and we state them because they rule out the simplest explanation. First, our 15 demonstrations average 1.60 tool calls, so the fine-tuned model at 1.13 has gone *below* the training distribution rather than converging to it. Second, on GSM8K mean tool calls did not fall at all (3.22 $\rightarrow$ 3.25). The effect is therefore not a global shortening of trace length, which would have affected both benchmarks. What the demonstrations have in common is that every search they contain returns a directly usable fact; none models recovering from an unhelpful observation. We interpret the effect as a reduced propensity to issue a *follow-up* search after an observation that does not already contain the answer—a search-specific behavior—rather than a learned length prior.

We term this failure mode **search-horizon truncation**. It is invisible to in-domain evaluation (where one or two calls suffice), and at $n=100$ it is not reliably detectable from accuracy alone: the same intervention that moves accuracy by an amount indistinguishable from noise moves trace depth by five orders of magnitude in p . This asymmetry is the practical lesson. We recommend that agent fine-tuning reports include tool-call distributions, not merely means, alongside accuracy.

Ablation: do longer demonstrations repair it? We expect that mixing longer multi-hop demonstrations into the training set repairs the effect, since the demonstrations currently contain no example of recovering from an unhelpful observation. We leave this to future work.

4.4 Replication under a different quantization kernel

The HotpotQA INT4 and INT4+QLoRA configurations were re-run in a fresh session, in separate processes, under bitsandbytes 0.50.1—a different kernel version from the original round. Accuracy reproduced to within one task (INT4 44% $\rightarrow$ 43%; INT4+QLoRA 38% $\rightarrow$ 38%) and mean tool calls reproduced to two decimal places (1.47 $\rightarrow$ 1.48; 1.13 $\rightarrow$ 1.13). The trace-depth analysis above uses this replication run, for which per-task tool-call counts and full agent traces are released.

This is a useful counterweight to Section 4.5: between two recent bitsandbytes releases under a fixed harness the agent’s behavior is stable, so the variance we observed earlier is attributable to the harness and kernel changing together rather than to quantization kernels being inherently unstable. We also record peak allocated memory during evaluation: 3.77 GB for INT4 against 2.21 GB of weights—the gap is activations and KV cache, and it is the peak figure that determines whether a deployment target fits.

214 4.5 Reproducibility: quantized-agent scores are environment-sensitive

215 In an earlier evaluation round conducted with an older bitsandbytes kernel and a slightly differ-
216 ent harness, the identical INT4 model scored 92% on a 25-task subset of our diagnostic suite
217 (multi_step: 60%), and QLoRA appeared to “recover” accuracy to 96%. After unifying the har-
218 ness and upgrading the quantization library, base INT4 scores 98% and the recovery effect disap-
219 pears. We report this deliberately: small quantized models operate close to capability thresholds on
220 multi-step tasks, so kernel-level numerical differences can move headline agentic scores by several
221 points. Pinned library versions and a single shared harness across all configurations are therefore
222 not optional hygiene but a validity requirement for quantized-agent research.

223 5 Limitations

224 Our study covers one model family (Phi-3-mini), one quantization method (bitsandbytes NF4) and
225 one hardware target (T4); we test no INT8 intermediate point and no alternative scheme such as
226 GPTQ [8] or AWQ, and the latency conclusions in particular will differ on GPUs with fused low-bit
227 kernels. The diagnostic suite is small (50 tasks), uses a closed 27-entry fact base and is saturated by
228 every configuration, so it functions as a format-compliance check rather than a capability measure-
229 ment; differences of ≤ 3 points on the 100-task benchmarks are within our resolution, and HotpotQA
230 inclusion match on span answers is more permissive than exact match.

231 The FP16 configuration was evaluated in a separate cloud session from the INT4 configurations.
232 Accuracy comparisons are unaffected—decoding is greedy and the harness byte-identical—but the
233 latency comparison is not concurrent and shared-tenancy GPUs vary with host load.

234 The QLoRA study uses a single minimal demonstration set (15 traces) and a single training seed; we
235 show that this regime suppresses follow-up search, not that QLoRA cannot help agents trained on
236 longer or task-matched traces. Our INT4 and INT4+QLoRA configurations also differ by the dtype
237 changes that `prepare_model_for_kbit_training` introduces (layer norms and the LM head are
238 upcast to fp32) as well as by the adapters. We report no zero-adapter control, so attribution to the
239 trained weights rests on the tool-call mechanism rather than an isolated ablation; our released harness
240 implements this control. Finally, the scaffold is intentionally minimal (ReAct, greedy decoding, two
241 tools) and results may differ under richer scaffolds with reflection or planning.

242 6 Broader impacts

243 Reducing the memory cost of capable tool-using agents supports on-device and on-premise deploy-
244 ment, where data need not leave the user’s hardware, and lowers the cost of agentic inference; the
245 same accessibility is dual-use, since cheaper agents are also cheaper to run at scale for unwanted au-
246 tomation. Our negative result is itself safety-relevant: we document a cheap fine-tuning regime that
247 leaves headline accuracy close to unchanged while measurably suppressing an agent’s willingness to
248 gather evidence before answering. An agent that stops searching early is not obviously broken from
249 the outside; it answers more confidently on less information. Practitioners adapting small agents on
250 demonstration data should measure tool-call depth, not accuracy alone.

251 7 Conclusion

252 A 3.8B-parameter model quantized to 4 bits retains essentially all of its tool-using agentic
253 capability—identical per-task outcomes on our diagnostic suite, no statistically detectable differ-
254 ence on two standard benchmarks—at 29% of the weight memory. Two caveats qualify the pic-
255 ture: on commodity GPUs quantization trades latency for memory rather than improving both, and
256 lightweight fine-tuning on short demonstration traces can silently suppress follow-up tool use, de-
257 grading exactly the multi-hop behavior agents exist to perform. Measuring trace depth, pinning
258 the numerical environment and using paired tests should be standard practice for efficient-agent
259 research.

260 **Reproducibility.** Code, harness, demonstrations, per-task CSVs and the analysis script are
261 anonymized at ANONYMIZED-REPO-URL.

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